

# A screened Rydberg formula for the successive ionization energies of all atoms and ions

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**\*\*Abstract—**We present a single closed-form expression, the *screened Rydberg formula*, for the successive ionization energy  $IE(Z, N)$  of any atom or ion. It needs only the nuclear charge  $Z$  and the ground electron configuration. The formula keeps the hydrogenic form  $Ry Z_{\text{eff}}^2/n^2$ . Its effective charge is split into two parts. The first is an exact, parameter-free first-order screening constant  $\sigma_1 = -n^2\Delta E_1$  from the  $1/Z$  perturbation expansion; it is a rational number for every configuration, and we tabulate it for  $N = 1$ –110. The second is a bounded, fitted higher-order remainder that depends on the ion charge in an Edlén-type form. A saturation construction guarantees  $Z - N + 1 \leq Z_{\text{eff}} \leq Z$  for any parameter values (given  $0 \leq \sigma_1 \leq N - 1$ , which holds for every configuration we use) and keeps the exact  $Z^2$  and  $Z$  coefficients of the  $1/Z$  series. One-electron Dirac, recoil, finite-nuclear-size and QED corrections are added without fitted parameters. The final model has 33 fitted global parameters and no per-element or per-ion parameters. On all 5847 successive ionization energies in the NIST Atomic Spectra Database ( $Z = 1$ –110), its mean absolute percentage error (MAPE) is 1.87 % (median 0.94 %). That is 7.52 % for neutral-atom first ionization energies and 0.0012 % for hydrogen-like ions; the H-like figure is consistent with, but not independent of, the QED theory behind the NIST values. The model was chosen by a pre-registered score that never used the blind extrapolation test (its interpolation splits do contain heavy elements). In a single blind extrapolation (fit  $Z \leq 54$ , predict  $Z = 55$ –110), it gives 6.60 % MAPE (median 1.62 %). The same test gives 11.6 % for Slater's rules, 13.6 % for a published parameter-free screened-hydrogenic model scored on the same rows, and 57.2 % and 170 % for two earlier fitted variants without the bound (Table 1). Extrapolation to heavy neutral atoms remains the weak point (21.8 % MAPE on their first ionization energies). In the blind fit Pb and Tl come out at about 1.2 and 1.7 eV, against 7.42 and 6.11 eV, and Lr comes out negative. With all data fitted, the formula still places oganesson (3.66 eV) far below radon (NIST 10.75 eV). All of these failures are reported, not corrected.

**Index Terms—** ionization energy; screening constants;  $1/Z$  expansion; screened hydrogenic model; isoelectronic sequences; Slater's rules; blind validation.

## 1. Introduction

The Bohr energy  $Ry Z^2/n^2$  is exact only for a non-relativistic one-electron ion. For every other atom and ion the ionization energy (IE) is controlled by electron–electron repulsion, which has no closed-form solution for  $N \geq 2$ . Two broad routes exist.

**Screening rules.** Slater [1] replaced  $Z$  by an effective charge  $Z - \sigma$ , with  $\sigma$  given by simple counting rules fitted by hand to atomic data. Clementi and Raimondi [2], [3] obtained  $\sigma$  from optimised self-consistent-field orbital exponents of neutral atoms. Both are designed for orbitals of neutral atoms, not for the full set of successive IEs. Scored as IE formulas on all NIST ions with a total-energy difference, Slater's rules give 11.8 % MAPE and Clementi–Raimondi 120 % (§4).

**Perturbation theory in  $1/Z$ .** Layzer [4], [5] showed that the non-relativistic energy of a fixed configuration is an asymptotic series  $E = Z^2 E_0 + Z E_1 + E_2 + \dots$ .  $E_0$  is hydrogenic and  $E_1$  is a rational combination of hydrogenic Slater integrals, which fixes the exact  $Z \rightarrow \infty$  limit of the screening constant. Higher orders need continuum sums [6], [7]. Along isoelectronic sequences, Edlén [8] systematised the smooth variation of screening with ion charge, using expansions in  $1/(\zeta + s)$ .  $Z$ -expansion codes with relativistic corrections were developed for selected sequences [9].

**Screened hydrogenic models (SHMs).** Starting with Mayer [10] and More [11], plasma-physics codes use screened hydrogenic energies with tabulated or fitted screening constants: l-splitting [12], [13], analytical potentials [14], [15], relativistic constants fitted by genetic algorithm [16], and a parameter-free, self-consistent  $(Z, N)$ -dependent screening [17]–[22]. Average-atom and kinetics codes value these for speed and coverage of all ions [23]–[25].

**Ab initio methods.** Koopmans' theorem [26],  $\Delta$ SCF Hartree–Fock and Kohn–Sham DFT [27], [28], correlated non-relativistic energies [29] and Dirac–Fock total energies for all ground configurations up to  $Z = 118$  [30] give more accurate numbers. They are numerical procedures, not formulas.

**The gap.** We found no single closed-form expression that (i) covers every ion of every element from the configuration alone, (ii) reproduces the exact large- $Z$  behaviour of the  $1/Z$  expansion, (iii) reports every fitted parameter, and (iv) is validated on all NIST successive IEs with held-out tests fixed in advance, including a blind extrapolation to heavier elements. The SHM papers we read report accuracy on their fitting data. [20], for example, reports selected sequences. Scored on our rows (§4.5), Mendoza et al.'s constants give 2.82 % MAPE on the 5011 rows they cover. This paper tries to fill that gap. It does not claim a new law of atomic physics; §5.1 states what is rediscovered.

### Contributions.

1. A table of the exact first-order screening constants  $\sigma_1$  for every NIST ground configuration,  $N = 1$ –110, as a parameter-free,  $Z \rightarrow \infty$ -exact counterpart to Slater's  $\sigma$  (§2.1, §4.6).
2. The screened Rydberg formula:  $\sigma_1$  plus a bounded Edlén-type remainder with 33 global parameters (§2.2–2.4).
3. A validation protocol with a pre-registered selection score and a single blind extrapolation test, applied identically to the final model, its ancestors and published baselines (§3, §4).
4. A head-to-head against two published SHMs, re-implemented and scored with the same scorer (§4.5): the parameter-free Kregar/Di Rocco model on all 5847 rows, and the fitted constants of Mendoza et al. (2011) on the 5011 rows they cover.

## 2. Theory

Atomic units are used in §2.1, with energies converted to eV via  $Ry = 13.6057$  eV.  $Z$  is the nuclear charge,  $N$  the number of electrons before ionization, and  $Z_a = Z - N + 1$  the charge seen by the departing electron.  $(n, l)$  is the removed subshell and  $k$  its occupancy.

### 2.1 Exact first-order screening from the $1/Z$ expansion

Scaling  $r \rightarrow \rho/Z$  turns the non-relativistic Hamiltonian into  $H = Z^2[H_0 + Z^{-1}V]$ , with  $H_0$  hydrogenic and  $V = \sum 1/\rho_{ij}$ . Rayleigh–Schrödinger perturbation theory in  $1/Z$  then gives, for a fixed configuration and term,  $E(Z, N) = Z^2E_0 + ZE_1 + E_2 + \dots$ , and the ionization energy is

$$IE(Z, N) = Z^2\Delta E_0 + Z\Delta E_1 + \Delta E_2 + \dots, \quad \Delta E_k = E_k(N-1) - E_k(N).$$

- **Zeroth order.**  $\Delta E_0 = 1/(2n^2)$ , which is exactly Bohr's formula.
- **First order.**  $E_1 = \langle V \rangle$  over the  $Z = 1$  hydrogenic state. Every hydrogenic radial integral  $F^k$  and  $G^k$  is an exact rational number, because each product  $P_a P_c$  is a polynomial times  $e^{-\beta r}$  with rational  $\beta$ . Examples:  $F^0(1s, 1s) = 5/8$ ,  $F^0(1s, 2s) = 17/81$ ,  $G^0(1s, 2s) = 16/729$ . We evaluate them in exact rational arithmetic and combine them with exact angular algebra.
- **Open shells.** For open subshells,  $E_1$  is evaluated for the Hund ground term by projecting onto the highest-weight  $(S, L)$  subspace. Where hydrogenic degeneracy couples configurations (the Layzer complex, e.g.  $1s^2 2s^2$  with  $1s^2 2p^2$ ),  $V$  is diagonalised within the complex. For two or more open subshells, a high-spin coupled term is used; this is the one approximation in  $\sigma_1$ .

Completing the square in the first two terms defines the first-order screening constant

$$IE \simeq \Delta E_0 (Z - \sigma_1)^2, \quad \sigma_1 = -\frac{\Delta E_1}{2\Delta E_0} = -n^2 \Delta E_1.$$

$\sigma_1$  is Layzer's  $Z \rightarrow \infty$  screening constant. It has no adjustable parameter and depends only on the configuration, not on  $Z$ . Two checks against known values:

- **He-like:**  $E_1(1s^2) = 5/8$ , so  $\sigma_1(\text{He}) = 0.6250$  (Slater: 0.30).
- **Li-like:**  $E_1(1s^2 2s) = 5965/5832 = 1.0228052 = 5/8 + 2 \cdot 17/81 - 16/729$ , so  $\Delta E_1 = -0.397805$  and  $\sigma_1 = 1.5912$ .
- **Be-like:** the complex value  $E_1 = 1.5592742$  agrees with Layzer's.

The exact series fails for near-neutral atoms (§4.6). The expansion parameter is effectively  $N/Z$ . For a neutral atom,  $Z - \sigma$  is only 1–3, so a 10 % error in  $\sigma$  becomes a 100–1000 % error in IE. A closed form for the higher orders does not exist:  $E_2$  already requires sums over the hydrogenic continuum. For example, the completed-square guess  $\Delta E_2 = \Delta E_1^2 / (4\Delta E_0)$  is 24–26 % too large for He- and Li-like ions. This is why the remainder in §2.2 has to be fitted.

## 2.2 The screened Rydberg form with a bounded remainder

The final model ("pa\_hier\_rel") is

$$\text{IE}(Z, N) = \mu(Z) \left\{ \text{Ry} \frac{Z_{\text{eff}}^2}{n^2} F_{n,j}(Z_{\text{eff}}) \left[ 1 + r_c \frac{(Z\alpha)^2}{n} \left( \frac{Z_{\text{eff}}}{Z_a} - 1 \right) \right] + \text{Ry} \frac{x_l K_l(k)}{n^2} \right\} - [\Delta E_{\text{QED}} + \Delta E_{\text{FNS}}]_{Z,n} \left( \frac{Z_{\text{eff}}}{Z} \right)^2,$$

where the last term applies only to the removal of an ns electron with  $n \leq 2$ , and

$$Z_{\text{eff}} = Z - \sigma_1(\mathcal{C}) - D, \quad D = \frac{T}{Z_a + \kappa + |T|/h}, \quad h = \begin{cases} (N-1) - \sigma_1 & T \geq 0 \\ \sigma_1 & T < 0 \end{cases}, \quad T = \sum_g \tau_g \nu_g + \sum_c \delta\tau_c \nu_c.$$

The terms are:

- **v\_g (five screening groups).** Let  $(n, l)$  be the removed subshell. Each of the other  $N - 1$  electrons, in a subshell  $(n', l')$ , is counted in exactly one group.
  - *same*: the  $k - 1$  other electrons of  $(n, l)$ .
  - *out*: outer electrons,  $n' > n$ , or  $n' = n$  with  $l' > l$ .
  - All other electrons ( $n' < n$ , or  $n' = n$  with  $l' < l$ ) are inner.
    - For an s or p target, inner electrons form *in* if  $n' \geq n - 1$  and *core* if  $n' \leq n - 2$ . *in* is the ns electrons of an np target plus the whole  $(n - 1)$  shell, including its d and f electrons.
    - For a d or f target, all inner electrons form *df*, from the same- $n$  lower- $l$  subshells down to 1s.

Hence  $\sum_g v_g = N - 1$ , and each group has a coefficient  $\tau_g$ . Examples, as  $(v_{\text{same}}, v_{\text{in}}, v_{\text{core}}, v_{\text{df}}, v_{\text{out}})$ :

- O  $2p^4$ : (3, 4, 0, 0, 0);
- Na 3s: (0, 8, 2, 0, 0);
- Fe  $3d^6 4s^2$ , 4s removed: (1, 14, 10, 0, 0), because  $3d^6$  belongs to the  $(n - 1)$  shell;
- Pb  $6p^2$ : (1, 20, 60, 0, 0), with  $6s^2 5s^2 5p^6 5d^{10}$  in *in* and every  $n' \leq 4$  electron, including  $4f^{14}$ , in *core*.
- **v\_c and  $\delta\tau_c$  (screening classes, Table A2).** The class of an electron depends only on the target type (s/p, d or f), on  $n - n'$  and on  $l'$ . Each class lies inside one group, and an electron of class  $c$  in group  $g$  contributes  $\tau_g + \delta\tau_c$  to  $T$ . The deviations are shrunk toward their group value by a ridge penalty ( $10^{-4}$  per row). Two of the 21 classes are empty in all 5847 rows and carry no  $\delta\tau_c$ : *sn\_out* ( $n' = n, l' > l$ ) and *out\_sp* ( $n' > n$ , s/p target). That leaves the 19 of Table A1. The bounded 9-parameter variant (pa\_bound9) and the pocket formula (Appendix A) use the same five groups without classes. An independent implementation of this rule reproduces the code's group and class counts with 0 mismatches on all 5847 rows and on all 7021 configurations with  $Z \leq 118$  (`tools/check_grouping_rule.py`).
- **Removed subshell.**  $(n, l)$  is the subshell whose occupancy drops from the  $N$ -electron configuration to the NIST ground configuration of the  $(N - 1)$ -electron ion. The  $N$ -electron configuration is the NIST ground configuration, or the Madelung order if the ion is not tabulated. For a rearranging ion (e.g.  $V 3d^3 4s^2 \rightarrow V^+ 3d^4$ ) it is the subshell that loses the most electrons (ties: larger  $n$ , then larger  $l$ ). If the  $(N - 1)$ -electron ion is not tabulated, or the configuration is supplied by the user, the outermost subshell (largest  $n$ , then largest  $l$ ) is removed.
- **$K_l(k) = [P(k) - P(k-1)] - 2l(k-1)/(4l+1)$**  is the change in the number of parallel-spin pairs relative to a statistical average (Hund kink).  $P(k)$  is the number of parallel-spin pairs of  $l^k$  under Hund's first rule. Equivalently,  $K_l(k) = (2l+1)(k-1)/(4l+1)$  for  $k \leq 2l+1$  and  $K_l(k) = -K_l(4l+3-k)$  above half filling. So:
  - $K_s \equiv 0$ ;
  - $K_p = 0, 0.6, 1.2, -1.2, -0.6, 0$  for  $k = 1 \dots 6$ ;
  - $K_d = (5/9)(0, 1, 2, 3, 4, -4, -3, -2, -1, 0)$ ;
  - $K_f = (7/13)(0, 1, \dots, 6, -6, \dots, -1, 0)$ .

The amplitude  $x_l$  is fitted for  $l = p, d, f$ . s electrons have no Hund term.

- **$\sigma_1$  as used.**  $\sigma_1$  is evaluated for the *frozen* configuration: the N-electron configuration minus one (n, l) electron, not the ground configuration of the ion. The two differ only for the 63 rearranged rows. It keeps  $\Delta E_0 = 1/(2n^2)$  on those rows.
- **$\mu(Z) = M/(M + m_e)$**  is the reduced-mass factor. M is the nuclear mass of the isotope used in the QED tabulation [31].
- **QED and finite-size term.** The QED and finite-nuclear-size shift is applied only to 1s and 2s removal (427 rows). It is the one-electron shift of §2.4 for charge Z, scaled by  $(Z_{\text{eff}}/Z)^2$ , and amounts to at most 0.9 % of the IE (at Z = 110, N = 2).
- **H-like ions.** For N = 1, T = 0, so D = 0.

**Reference implementation.** A reader cannot reconstruct  $\sigma_1$  (frozen configuration),  $\mu(Z)$  or the QED/FNS shift from the text alone. We therefore publish every per-row input in `results/model_inputs.csv`: removed subshell, k, j,  $\sigma_1$ ,  $v_g$ ,  $v_c$ , K,  $\mu$ , QED/FNS. A short script that uses only this table, the equation above and the parameters of Table A1 (`tools/verify_from_inputs.py`, which imports no model code) reproduces the production code on all 5847 rows to  $7 \cdot 10^{-16}$  relative, both for the final model and for `pa_bound9`.

The saturation form of D has two properties that hold for any parameter values.

1. **The bound  $Z_a \leq Z_{\text{eff}} \leq Z$ .** Write  $D = \text{sign}(T) \cdot h \cdot u / (1+u)$  with  $u = |T|/[h(Z_a + \kappa)]$ . Then  $|D| < h$ , so  $Z_{\text{eff}}$  can neither fall below the fully screened, non-penetrating limit  $Z_a$  nor exceed the bare charge Z. Since  $h = (N-1) - \sigma_1$  or  $\sigma_1$ , this requires  $0 \leq \sigma_1 \leq N - 1$ . That is a checked property of the configurations, not a proven one: it holds for all 7021 ground/Madelung configurations with  $Z \leq 118$ , with no violations, and the code warns if a user-supplied configuration breaks it.
2. **The exact asymptotics.** At fixed N and  $Z_a \rightarrow \infty$ ,  $D \rightarrow T/(Z_a + \kappa) = O(1/Z)$ . The expansion of IE therefore begins  $Ry[Z^2 - 2Z\sigma_1 + O(1)]/n^2$ , and its  $Z^2$  and Z coefficients are those of the exact  $1/Z$  series. The fitted part only represents  $\Delta E_2$  and higher orders: relaxation, correlation and penetration at low ion charge.

## 2.3 Charge-dependent penetration as an Edlén-type term

The remainder's dependence on  $1/(Z_a + \kappa)$  was found empirically (`docs/semi_empirical.md`). Along every isoelectronic sequence, the excess charge  $p = Z_{\text{eff}} - Z_a$  grows with ion charge q as  $p \propto -\tau/(Z_a + \kappa)$ , with one  $\kappa$  common to all sequences. For the Na sequence (3s), for example,  $p = 0.84, 1.15, 1.34, 1.46, 1.56$  for  $q = 0-4$  and 2.11 at  $q = 20$ . Introducing this term reduced the error of the purely fitted model from about 9 % to about 2 % MAPE. This is an independent rediscovery of the isoelectronic regularity that Edlén formalised [8]. It is also consistent with Layzer's theory, in which the screening constant is  $\sigma_0 + \sigma_1'/Z + \dots$ . We do not present it as a new law. What is specific here is narrower: one universal denominator across all sequences, combined with exact  $\sigma_1$  and with the saturation bound of §2.2.

## 2.4 Relativistic, recoil, finite-size and QED corrections

- **$F_{\{n,j\}}(\zeta)$**  is the exact ratio of the point-nucleus Dirac binding energy to the Schrödinger energy for charge  $\zeta$ . With  $x = \zeta\alpha$  and  $\kappa = j + 1/2$ ,  $F = (2n^2/x^2) \cdot \{1 - [1 + (x/(n - \kappa + \sqrt{\kappa^2 - x^2}))^2]^{-1/2}\}$ . j is assigned by jj filling:  $l - 1/2$  while  $k \leq 2l$ , otherwise  $l + 1/2$ . (We write F to avoid confusion with the screening remainder D.)
- **The bracket  $1 + r_c(Z\alpha)^2(Z_{\text{eff}}/Z_a - 1)/n$**  is a Fermi-Segrè-type correction for the penetration of the valence electron into the region of full nuclear charge. Its five coefficients  $r_c$  (classes s,  $p^{1/2}$ ,  $p^{3/2}$ , d, f) are fitted.
- **One-electron terms.** For one-electron ions we use the Dirac energy with a numerically solved finite-nuclear-size (FNS) shift, Barker-Glover recoil, the Uehling vacuum polarisation computed over the Dirac 1s density, and the one-loop self-energy function  $F_{\text{SE}}(Z\alpha)$  from the all-order tabulation of Yerokhin and Shabaev [31]. The  $2 \times 110$  tabulated values are external theory inputs, not fitted parameters. In the many-electron formula, the QED and FNS shifts are applied only to 1s and 2s removal, scaled by  $(Z_{\text{eff}}/Z)^2$ .
- **No closed-form substitute for the table.** The closed-form low-order  $Z\alpha$  expansion of  $F_{\text{SE}}$  matches the table at  $Z = 1$  but diverges for  $Z \gtrsim 15$ . It gives 0.354 % MAPE on H-like ions, worse than no QED.

2.5 Density functional theory as a physics check

We also wrote a radial Kohn–Sham LSDA solver (Slater exchange + VWN5 correlation). It reproduces the NIST LDA reference total energies [28] to about  $10^{-6}$  hartree (e.g. Ne  $-128.233481$ ). Because that reference uses the same functional, the agreement verifies the implementation only, not physical accuracy.  $\Delta$ SCF ionization energies from this solver have no fitted parameters and serve as an independent, non-formula check (§4.2). DFT does not enter the formula.

2.6 Parameter count

The final model has 33 fitted global parameters:

| block                                                           | count |
|-----------------------------------------------------------------|-------|
| group screening $\tau_g$                                        | 5     |
| saturation denominator $\kappa$                                 | 1     |
| class deviations $\delta\tau_c$ (ridge-shrunk; counted in full) | 19    |
| relativistic penetration $r_c$                                  | 5     |
| Hund amplitudes $x_l$ (p, d, f)                                 | 3     |

There are no per-element or per-ion parameters, and the prediction code never reads a NIST ionization energy. Fitted values are in Table A1 (Appendix) and `results/uni_final_params.json`, and the class definitions are in Table A2.

**Model names used in this paper.** Each name refers to exactly one model.

| name                                                                         | params | definition                                                                                        |
|------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------|
| <b>Screened Rydberg formula</b> (final; code name <code>pa_hier_rel</code> ) | 33     | eq. (2.2) with groups, classes, relativistic bracket                                              |
| <b>bounded 9-parameter variant</b> ( <code>pa_bound9</code> )                | 9      | eq. (2.2) with the five groups only (no $\delta\tau_c$ ) and no relativistic bracket ( $R = 1$ )  |
| <b>pocket formula</b> ( <code>uni_pocket</code> )                            | 8      | Appendix A. No $\sigma_1$ , no bound: $Z_{\text{eff}} = Z - \sum s_g v_g - t(N-1)/(Z_a + \kappa)$ |

`pa_bound9` uses the pocket formula's five electron groups but is *not* the pocket formula.

3. Data and validation protocol

3.1 Data

The reference data are the 5847 successive ionization energies of the NIST Atomic Spectra Database [32] for  $Z = 1$ –110, all charge states, with NIST ground configurations (`data/nist_ie.csv`). The database flags each value as experimental (311 rows), semi-empirical (919) or theoretical (4617). Every metric below is computed over all 5847 rows unless stated otherwise. By status, the final model's all-data fit gives (`evaluate.py`, `status= strata`): experimental 4.63 % MAPE (median 2.86 %,  $n = 311$ ), semi-empirical 2.20 % (median 0.83 %,  $n = 919$ ), theoretical 1.62 % (median 0.92 %,  $n = 4617$ ). The experimental rows are mostly neutral atoms and low-charge ions (98 neutral, 193 of 311 with charge  $\leq 2$ , median charge 2), the hardest regime for the formula, so their higher error mirrors the neutral-atom weakness rather than a disagreement with experiment specifically. The data were taken from NIST ASD version 5.12 [32], the current version at the time of access, and accessed on or before 5 October 2026. The project log records results computed from these data on that date; the original download timestamp was not preserved.

Subsets used below:

- 108 neutral-atom first IEs;
- 110 hydrogen-like ions;
- 63 "rearranged" ions, whose ground configuration changes by more than one electron on ionization (e.g.  $V\ 3d^3 4s^2 \rightarrow V^+ 3d^4$ ).

The configuration of each ion is an input. Inside the table it is the NIST ground configuration. For ions not in the table the Madelung order is used: all ions with  $Z > 110$ , and 258 ions with  $Z = 104$ –110 that are missing from the table. They enter no fit and no score, only the 7021-configuration checks (§4.7) and predictions outside the table.

The error measure is  $\text{MAPE} = (100/M) \sum |\text{IE}_{\text{pred}} - \text{IE}_{\text{NIST}}| / \text{IE}_{\text{NIST}}$ , together with the median absolute percentage error. All scores come from one shared scorer (`evaluate.py`). Fits minimise squared log-ratios  $\ln(\text{IE}_{\text{pred}}/\text{IE}_{\text{NIST}})$ , so a 4 eV and a 100 keV ionization energy carry equal relative weight.

### 3.2 Held-out splits

Every fitted model is refit on the training rows of each split with its own fitting routine:

| split | training rows      | test / validation rows                           | n test rows | role                         |
|-------|--------------------|--------------------------------------------------|-------------|------------------------------|
| V1    | $Z \leq 36$        | $37 \leq Z \leq 54$                              | 819         | selection                    |
| V2    | $Z \leq 44$        | $45 \leq Z \leq 54$                              | 495         | selection                    |
| S1    | $Z \bmod 5 \neq 0$ | $Z \bmod 5 = 0$ (unseen elements)                | 1188        | selection                    |
| S3    | $N \bmod 6 \neq 0$ | $N \bmod 6 = 0$ (unseen isoelectronic sequences) | 928         | selection                    |
| S2    | $Z \leq 54$        | $Z \geq 55$                                      | 4362        | <b>blind</b> , reported once |

**Selection score.** The score is the mean MAPE over V1, V2, S1 and S3. It was fixed before the final round of model development and was the only quantity used to choose the final model. S2 (fit  $Z \leq 54 \rightarrow Z \geq 55$ ) was never used for any choice: it was computed once per candidate, after its specification was frozen. The selection splits themselves are not free of heavy elements, however. As pre-registered, S1 and S3 contain  $Z \geq 55$  rows in both training and test sets (913 of the 1188 S1 test rows and 703 of the 928 S3 test rows, about 76 %), so heavy-atom *interpolation* accuracy did inform the design choices; only heavy-atom *extrapolation* was kept blind. As a robustness check (reporting only, after the audit), we recomputed the selection score with S1 and S3 restricted to  $Z \leq 54$  in both training and test. The winner is unchanged: `pa_hier_rel` 1.844, then the 28-parameter variant without the relativistic term 1.989, the bounded 9-parameter variant 2.094 and the lowest-scoring Push B candidate 2.118. This checks the ranking of the frozen candidates, not the exploration path that produced them. The final model's numbers were reproduced by an independent validation script, which matched the developing agent's own run with a difference of 0.0 in every split.

### 3.3 History of the protocol (disclosure)

The protocol above was the third. In earlier stages of this project:

1. A purely fitted model reported S2 = 19 % using a bound on core screening chosen after seeing S2. Without that bound, the blind value is 170 %.
2. Later, an intermediate unified model was chosen on a score that included S2.

Both are reported here only as non-blind history (§4.3). Other disclosures:

- The developers of the final round knew which heavy near-neutral atoms the previous model failed on, and that an 8-parameter model extrapolated well. This motivated the direction of the search, though not the choice, which was made on the selection score.
- About 57 variants were explored in that round (43 logged by one agent, about 14 by the other). The minimum selection score over them is therefore optimistic.
- The screening classes and the Hund term were designed earlier on all rows, including  $Z \geq 55$ . This is structural leakage that we cannot remove retrospectively.
- **A later round of four pre-registered variants** was tried after this paper's first draft (`docs/preregistration_v2.md`, `models/v2/NOTES.md`): a relativistic factor that cannot change sign, a second-order charge term, and a  $\kappa$  per orbital type. The non-negative relativistic factor fixes the signs of `Lr` and `Og`, but none of the variants improved the selection score ( $2.15 \rightarrow 2.18\text{--}2.49$ ). The model was therefore left unchanged. This brings the total number of explored variants to about 61. The pre-registration file was fingerprinted (`sha256`, `models/v2/PREREG_HASH.txt`) before round 1 and committed to git afterwards.

### 3.4 Computational reproducibility

The frozen run used Python 3.13 with NumPy and SciPy; their exact versions were not recorded. We repeated every refit in an independent software environment (Python 3.11.9, NumPy 2.4.4, SciPy 1.17.1; `docs/review/reproducibility_py311.md`).

- **Final model:** reproduced to within 0.005 percentage points in every split. Selection score 2.1473 vs 2.1481; blind S2 6.6030 vs 6.6031 %; all data 1.8742 % in both.
- **Unbounded reference models:** these depend on the least-squares path.
  - The pocket formula's V1 fit, which diverges in the frozen run (MAPE  $4.4 \cdot 10^6$  %), converges in the second environment to 4.35 % (selection score 4.19).
  - The u29 V1 fit does the opposite: it converges in the frozen run and diverges in the second.
  - u35's blind S2 moves from 57.2 % to 56.5 %.

Their V1 entries in Table 1 therefore describe the optimizer as much as the model. We do not use the pocket formula's V1 divergence as evidence against it.

## 4. Results

### 4.1 Baselines versus the final model

All cross-model numbers in this section come from one generated matrix with the row count in every cell, `results/benchmark_matrix.md` (`tools/benchmark_matrix.py`). Comparisons between models always use the same rows.

**Table 1.** All values are MAPE in %. "All", "neutral" and "H-like" come from all-data fits. V1, V2, S1, S3 and S2 are held-out values after refitting. For parameter-free models nothing is fitted, so the split columns are simply the error on those rows. Sources: `results/model_comparison.md`, `results/uni_validation.md`.

| model                                                      | fitted<br>params | all<br>(median)        | neutral<br>1st IE | H-like        | V1                           | V2          | S1          | S3          | selection<br>score | blind S2<br>(median /<br>neutral) |
|------------------------------------------------------------|------------------|------------------------|-------------------|---------------|------------------------------|-------------|-------------|-------------|--------------------|-----------------------------------|
| Bohr Ry $Z^2/n^2$                                          | 0                | 1363<br>(172)          | $2.3 \cdot 10^4$  | 6.00          | 1050                         | 1045        | 1317        | 1399        | 1203               | 1517                              |
| Slater 1930, total-energy<br>difference                    | 0                | 11.8 (7.48)            | 51.8              | 6.00          | 13.4                         | 14.4        | 11.6        | 12.4        | 12.9               | 11.6 (8.07 /<br>50.7)             |
| Clementi–Raimondi, total-<br>energy difference             | 0                | 120 (18.7)             | 2281              | 6.00          | 87.1                         | 77.3        | 116         | 119         | 99.9               | 134                               |
| Exact 1/Z series, completed<br>square + rel/QED ("zexp")   | 0                | 69.8 (7.20)            | 1105              | 0.0012        | 53.4                         | 54.0        | 68.3        | 74.3        | 62.5               | 77.8 (– / 1772)                   |
| GSHM final (purely fitted,<br>blind)                       | 32               | 1.57 (0.74)            | 7.61              | 0.19          | 34.1                         | 1.65        | 1.59        | 1.70        | 9.77               | 170 (3.51 /<br>7393)              |
| exact $\sigma_1$ + linear remainder<br>("u35")             | 35               | 1.48 (0.73)            | 6.43              | 0.0012        | 26.4                         | 1.48        | 1.52        | 1.65        | 7.76               | 57.2 (2.00 /<br>3831)             |
| pocket formula (hand-<br>calculable)                       | 8                | 4.68 (3.04)            | 16.8              | 1.16          | $4.4 \cdot 10^6$ or<br>4.35† | 2.55        | 4.62        | 5.24        | – or 4.19†         | 11.3 (2.04 /<br>12.3)             |
| bounded 9-parameter variant<br>(pa_bound9)                 | 9                | 2.90 (1.39)            | 12.0              | 0.0012        | 2.16                         | 1.95        | 2.90        | 3.05        | 2.51               | 7.80 (– / 28.0)                   |
| <b>Screened Rydberg formula<br/>(final, "pa_hier_rel")</b> | <b>33</b>        | <b>1.87<br/>(0.94)</b> | <b>7.52</b>       | <b>0.0012</b> | <b>3.09</b>                  | <b>1.60</b> | <b>1.91</b> | <b>2.00</b> | <b>2.15</b>        | <b>6.60 (1.62 /<br/>21.8)</b>     |

† Optimizer-path dependent (§3.4): the V1 fit diverges in the frozen run and converges to 4.35 % in a second software environment.

**How much better than Slater.** The ratios below are from `results/benchmark_matrix.md`.

- All-row mean MAPE: Slater's rules (11.8 %) are 6.3× less accurate than the final model (1.87 %) and 4.1× less accurate than pa\_bound9 (2.90 %).
- Median: 7.48 % against 0.94 %.
- Blind S2: 11.6 % against 6.60 %, a factor of 1.8.

The final model has the lowest selection score of all candidates (2.15). The next were the other developer's choice, "pb\_clip\_pos" (30 parameters), at 2.37 and the 28-parameter hierarchical variant "pa\_hier" at 2.37 (blind S2 13.9 %

and 8.47 %, respectively). In-distribution, the bound costs a little accuracy: 1.87 % against 1.48 % for u35, and 7.52 % against 6.43 % on neutral atoms. In return, it removes the catastrophic extrapolation failures.

### 4.2 Errors by stratum

**Table 2.** All-data fit, MAPE in % (source: `results/lit_comparison.md` / `docs/literature.md` §2.2).

| model                | $N \leq 10$  | $11 \leq N \leq 36$ | $N \geq 37$ | $Z \geq 55$ | removed d   | removed f   |
|----------------------|--------------|---------------------|-------------|-------------|-------------|-------------|
| Slater 1930          | 4.89         | 9.38                | 16.8        | 11.6        | 16.2        | 12.9        |
| pocket formula (8 p) | 2.37         | 2.76                | 7.38        | 4.7         | 3.67        | 11.5        |
| GSHM final (32 p)    | 0.698        | 1.16                | 2.31        | 1.48        | 0.982       | 2.67        |
| u35 (35 p)           | 0.463        | 1.08                | 2.28        | 1.58        | 1.17        | 2.58        |
| <b>final (33 p)</b>  | <b>0.439</b> | <b>1.43</b>         | <b>2.87</b> | <b>1.92</b> | <b>1.52</b> | <b>3.51</b> |

- **Charge.** The error grows with the number of electrons and falls with ion charge. On ions of charge  $\geq 3$  the final model gives 1.61 %.
- **Where theory alone suffices.** For few-electron, highly charged ions, the parameter-free exact series is already good: median 0.7 % for  $N \leq 10$ , and 0.48 % MAPE for  $N/Z \leq 0.2$  with two terms.
- **Neutral atoms, DFT check.** The parameter-free LSDA  $\Delta$ SCF solver reaches 3.30 % on the 54 neutral first IEs with  $Z \leq 54$ , and 0.80 % on all 171 ions with  $Z \leq 18$ . It is better than every closed form on neutral atoms, but it is a numerical procedure and was run on only 207 rows.

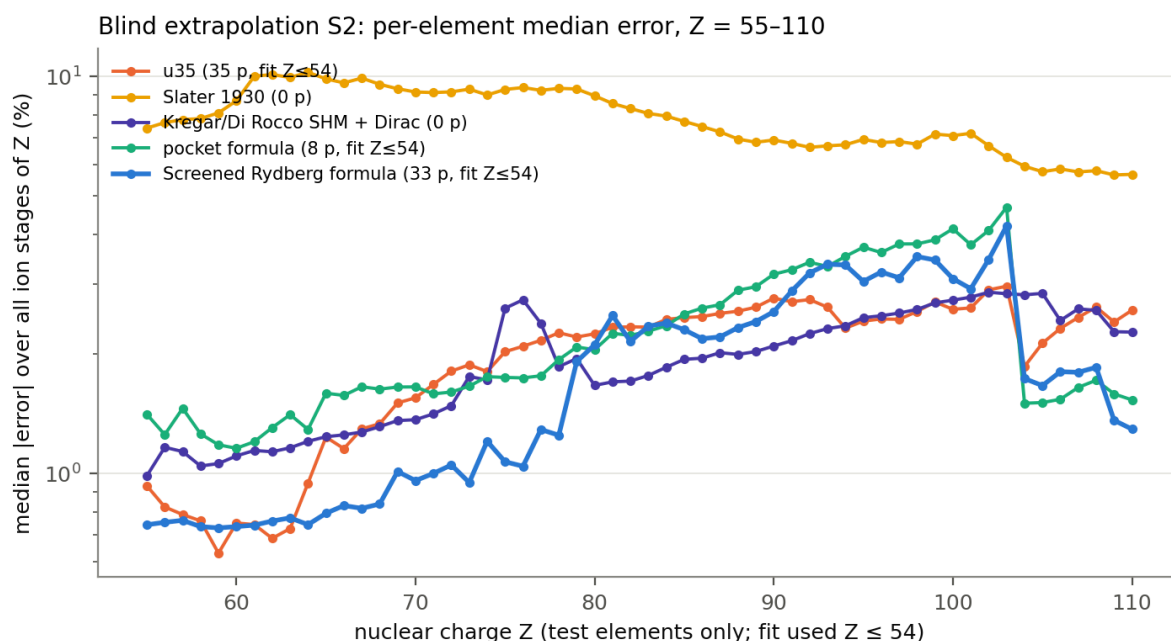
Figures (existing): parity plot `results/figures/uni_parity.png`; first IEs against Z `results/figures/uni_first_IE.png`; successive IEs of selected elements `results/figures/uni_successive.png`; residuals `results/figures/uni_residuals.png`.

### 4.3 Blind extrapolation (S2)

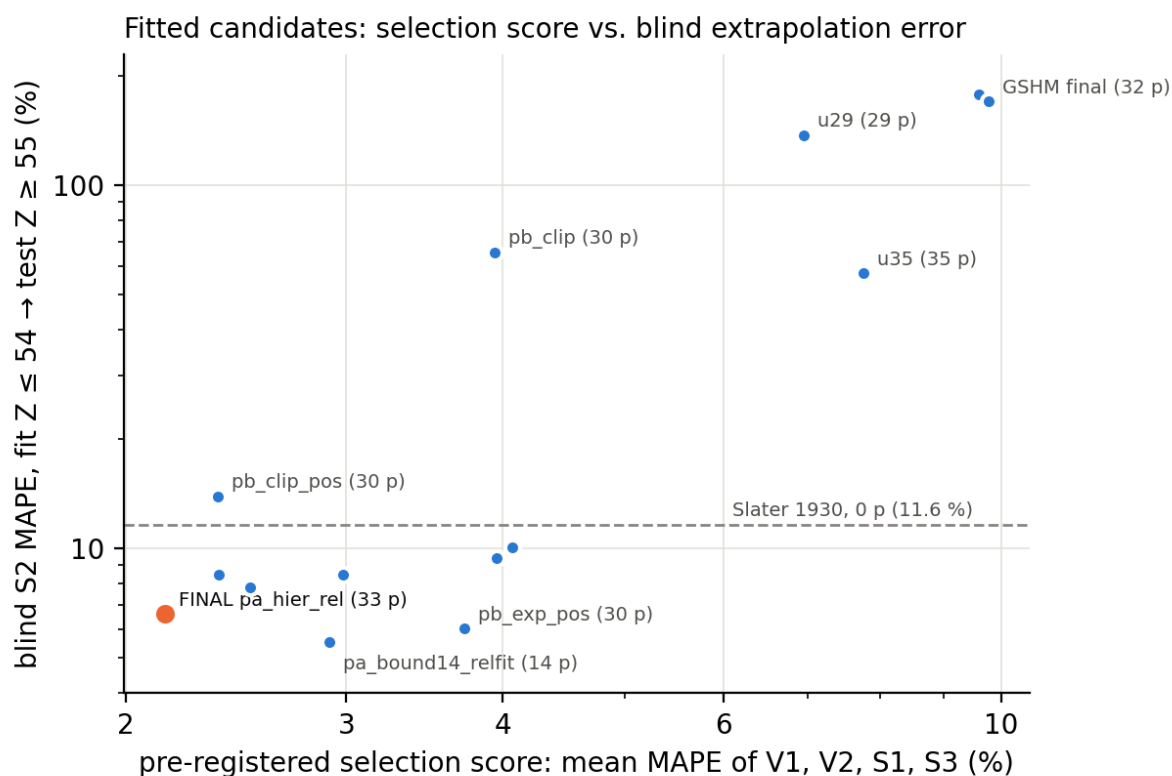
With a fit on  $Z \leq 54$  only, the final model predicts the 4362 ionization energies of  $Z = 55\text{--}110$  with 6.60 % MAPE and a 1.62 % median. The ancestors of the formula show what each ingredient contributes:

| model                                               | what it adds                                                                 | blind S2 MAPE (%) | S2 neutral (%) |
|-----------------------------------------------------|------------------------------------------------------------------------------|-------------------|----------------|
| GSHM final (fitted screening, no exact $\sigma_1$ ) | –                                                                            | 170               | 7393           |
| u35                                                 | exact $\sigma_1$ as the large-Z anchor, linear remainder                     | 57.2              | 3831           |
| final                                               | + saturation bound $Z_a \leq Z_{\text{eff}} \leq Z$ , hierarchical shrinkage | <b>6.60</b>       | <b>21.8</b>    |
| Slater 1930 (0 p, nothing extrapolated)             | –                                                                            | 11.6              | 50.7           |

- **What fixed extrapolation.** Exact  $\sigma_1$  alone was not enough. The decisive step was confining the fitted part to a bounded remainder. In a 14-parameter ablation, the bound improved the selection score from 3.96 to 2.91; its blind S2 changed from 9.43 % to 5.53 %, and that S2 change was seen only after the choice.
- **Ions versus neutral atoms.** For ions ( $Z > N$ ) the blind S2 MAPE is 6.4 %. For neutral atoms it is 21.8 %.
- **Per element.** Figure 1 shows the per-element median error on the S2 test rows.
- **Does the selection score predict extrapolation?** Figure 2 plots selection score against blind S2 for all fitted candidates. Candidates with a low selection score generally extrapolate well, but the relation is not monotone. A candidate with a worse selection score (pb\_exp\_pos, 3.73) has a lower blind S2 (6.03 %) than the final model. It was not promoted, because doing so would be selection on S2.



**Figure 1.** Blind extrapolation. Each point is the median absolute percentage error over all ion stages of one element  $Z = 55\text{--}110$ . The three fitted models are fitted on  $Z \leq 54$  only; Slater's rules and the Kregar/Di Rocco SHM have no fitted parameters. Generated from refits with the project's validation code; the S2 MAPEs reproduce Table 1 (final 6.603 %, u35 57.233 %, pocket 11.334 %, Slater 11.638 %).



**Figure 2.** Selection score (mean of V1, V2, S1, S3; no S2 data, although S1 and S3 contain  $Z \geq 55$  rows) against blind S2 MAPE for every fitted candidate in `results/model_comparison.csv`. The pocket formula is omitted because its V1 fit diverged in the frozen run. That divergence depends on the optimizer path (§3.4); with the converged V1 its selection score would be 4.19. The dashed line is Slater's rules (0 parameters).

**Failures of the blind fit**, all reported here and none corrected, since correcting them now would be post hoc:

- **Heavy p-block neutrals.** Their first IEs are badly underestimated (`results/known_failures.json`, refit on  $Z \leq 54$ ):

- Pb: 1.20 eV against 7.42 eV;
- Tl: 1.74 against 6.11;
- Rn: 6.10 against 10.75.

The frozen run gave 1.19 and 1.73; refits in the two software environments differ by about 0.01 eV here.

- **Lr is negative.** Lr is predicted at -3.33 eV. The fitted  $r_c$  are negative, and the relativistic bracket is not bounded, so the  $Z_{\text{eff}}$  bound alone does not guarantee a positive IE once  $Z\alpha$  is large.
- **f removal.** Third IEs of lanthanides and actinides (4f/5f removal) are about 2.0–2.5 times too high (mean 2.2, median 2.1, 23 rows; worst  $\text{Er}^{2+}$  at 55.65 eV against 22.7 eV), because the  $Z \leq 54$  training set contains no f electrons.

## 4.4 Hydrogen-like ions: correction layers

**Table 3.** 110 H-like ions,  $Z = 1\text{--}110$  (`docs/first_principles.md` §2.1).

| layer                                                                      | MAPE (%) | median (%) | max (%) |
|----------------------------------------------------------------------------|----------|------------|---------|
| Bohr Ry $Z^2$                                                              | 6.0      | 4.2        | 19.5    |
| (a) Dirac, point nucleus                                                   | 0.190    | 0.118      | 0.91    |
| (b) + recoil + finite nuclear size                                         | 0.113    | 0.109      | 0.248   |
| (c) + one-loop QED (Uehling computed; $F_{\text{SE}}$ from [31])           | 0.00117  | 0.000154   | 0.0096  |
| (d) as (c) but with the closed-form $Z\alpha$ expansion of $F_{\text{SE}}$ | 0.354    | 0.191      | 1.18    |

**Caveat.** The NIST H-like reference values are themselves computed from the same QED theory. Layer (c) therefore shows consistency with that source, not independent validation. The remaining  $10^{-5}\text{--}10^{-4}$  residual at high  $Z$  is the omitted two-loop QED, nuclear-polarisation and recoil-QED terms.

## 4.5 Head-to-head with published screened hydrogenic models

Two published SHMs could be fully specified from articles we were able to read:

- the parameter-free Kregar/Di Rocco model [20], [22], implemented from its definitions;
- the relativistic model of Mendoza et al. [16], implemented from its published  $19 \times 19$  table of fitted screening constants.

The constants of the other two candidates, More [11] and Faussurier et al. [12] are in papers we could not access, and no verifiable reprint of the tables was found. We did not reconstruct them from memory.

### 4.5.1 Kregar/Di Rocco SHM (parameter-free; all 5847 rows)

We implemented it from its definitions:

- screening from hydrogenic densities with an exchange correction, iterated to self-consistency;
- $E = -\sum q_i Z_i^2 / 2n_i^2$ ;
- non-relativistic, Pauli and Dirac variants.

#### Fidelity of our Kregar/Di Rocco implementation.

- **Matches the published numbers:** same-shell  $Z \rightarrow \infty$  screening constants reproduce every printed digit (e.g. 1s 0.3125, 2p 0.3492), and total energies agree with the published table within 0.71 %.
- **Differs:** cross-shell constants differ by 0.011 on average (at most 0.053). The original uses fitted closed forms whose coefficients we could not obtain [19]. Valence IEs of near-neutral ions come out 3–5 eV higher than the printed model values; Ar I, for example, is 18.96 eV here against 14.72 eV printed.

Our near-neutral numbers are therefore "the model as defined", not the authors' code.

**Table 4.** Same 5847 rows, same scorer, MAPE in % (`results/lit_comparison.md`). For the 0-parameter models the S2 column is the error on the  $Z \geq 55$  rows. For the fitted models it is the blind S2 refit.

| model                                 | params    | all (median)       | neutral 1st IE | H-like        | charge $\geq 3$ | S2 ( $Z \geq 55$ ) |
|---------------------------------------|-----------|--------------------|----------------|---------------|-----------------|--------------------|
| Kregar/Di Rocco SHM, non-relativistic | 0         | 13.6 (2.34)        | 230            | 6             | –               | 15.1               |
| Kregar/Di Rocco SHM + Pauli           | 0         | 12.6 (1.79)        | 225            | 1.16          | –               | 13.9               |
| Kregar/Di Rocco SHM + Dirac           | 0         | 12.5 (1.71)        | 224            | 0.19          | 4.68            | 13.6               |
| Slater 1930                           | 0         | 11.8 (7.48)        | 51.8           | 6             | 10.2            | 11.6               |
| pocket formula                        | 8         | 4.68 (3.04)        | 16.8           | 1.16          | –               | 11.3               |
| <b>final</b>                          | <b>33</b> | <b>1.87 (0.94)</b> | <b>7.52</b>    | <b>0.0012</b> | <b>1.61</b>     | <b>6.6</b>         |

The 224 % neutral figure is for our implementation, built from the published definitions. It is not fully faithful for valence shells: it gives 18.96 eV for Ar I where the authors print 14.72 eV, and the authors' printed valence IEs are 3–5 eV lower than ours, so this figure probably overstates the error of the authors' own model on neutral atoms.

The parameter-free SHM works well for ionised species, with a median of 1.54 % for charge  $\geq 3$ . It fails near neutrality: 37 of its 5847 predictions are zero or negative, all for ions of charge 0–4 with an open d or f shell. The fair conclusion has three parts:

1. Where the  $Z \rightarrow \infty$  limit dominates, parameter-free and exact- $\sigma_1$  models both work.
2. Near neutrality, every parameter-free hydrogenic model degrades. The fitted remainder is what makes the formula useful there, at the price of 8–33 parameters.
3. The bounded fitted formula extrapolates better (6.6 %) than the parameter-free SHM scores on the same  $Z \geq 55$  rows (13.6 %). The unbounded fitted models extrapolate worse (57 % and 170 %).

#### 4.5.2 Mendoza et al. 2011 (published constants; 5011 covered rows)

*Source and validation.*

- The relativistic nlj screening constants of [16] ( $19 \times 19$  matrix  $\sigma_{kk'}$ , subshells  $1s_{1/2}$  to  $5p_{3/2}$ ) were transcribed from the authors' open-access deposit of the article (<https://oa.upm.es/11165/>), Tables 1 and 2.
- We implemented the model as published: Dirac energies of screened charges,  $Q_k = Z - \sum_{k'} \sigma_{kk'}(P_{k'} - \delta_{kk'})$ , and  $IE = E_T(N-1) - E_T(N)$  with NIST ground configurations.
- It reproduces six of the paper's printed tables to their rounding: 84 IEs in Table 3 to  $\leq 0.023$  %, and Tables 4–8 to  $\leq 0.16$  % (`models/benchmarks/mendoza2011/`).
- No parameter was fitted by us.

*Caveats.*

- The constants were fitted by the authors with a genetic algorithm to NIST and FAC energies of isoelectronic sequences from He to Eu, with  $Z$  up to 92. That overlaps our test rows, so no split is held out for this model.
- Neutral atoms and singly charged ions were excluded from their fit.
- The constants were optimised for ionization *and excitation* energies together, so an IE-only score is not the quantity they were optimised for.
- The tables stop at  $5p_{3/2}$ , so 836 rows ( $Z \geq 55$ , ground configurations with 5d, 5f, 6s, 6p, 6d or 7s electrons) have no prediction.

The comparison is therefore restricted to the **5011 covered rows**.

**Table 4a.** The 5011 rows covered by Mendoza et al.; same scorer, all three models fitted to all data (in-sample). Cells: mean / median absolute percentage error [n]. Source: `tools/compare_mendoza.py`  $\rightarrow$  `results/compare_mendoza.md`.

| rows                  | Screened Rydberg (final, 33 p) | bounded 9-parameter (pa_bound9) | Mendoza et al. 2011 |
|-----------------------|--------------------------------|---------------------------------|---------------------|
| all covered rows      | <b>1.62 / 0.78</b> [5011]      | 2.63 / 1.19                     | 2.82 / 0.85         |
| ions only ( $Z > N$ ) | <b>1.55 / 0.76</b> [4957]      | 2.54 / 1.17                     | 2.56 / 0.82         |
| charge $\geq 3$       | <b>1.45 / 0.73</b> [4837]      | 2.43 / 1.14                     | 2.27 / 0.77         |
| $N \leq 10$           | 0.44 / 0.23 [1048]             | 0.51 / 0.30                     | <b>0.40 / 0.17</b>  |
| $11 \leq N \leq 36$   | <b>1.43</b> / 0.82 [2275]      | 2.12 / 1.17                     | 1.82 / <b>0.61</b>  |
| $N \geq 37$           | <b>2.62 / 1.51</b> [1688]      | 4.65 / 3.52                     | 5.68 / 2.48         |

| rows                                        | Screened Rydberg (final, 33 p) | bounded 9-parameter (pa_bound9) | Mendoza et al. 2011 |
|---------------------------------------------|--------------------------------|---------------------------------|---------------------|
| Z ≥ 55                                      | 1.57 / 0.72 [3526]             | 2.30 / 1.08                     | 2.74 / 1.00         |
| H-like                                      | 0.0012 [110]                   | 0.0012                          | 0.19                |
| experimental                                | 4.37 [236]                     | 7.17                            | 11.3                |
| neutral atoms (outside Mendoza's fit range) | 8.37 [54]                      | 11.5                            | 27.1                |
| 99th percentile / max APE                   | 11.3 / 30.4                    | 14.8 / 38.0                     | 28.4 / 119          |
| rows within 5 %                             | 92.2 %                         | 83.8 %                          | 86.4 %              |

**Summary.** On the ions it covers, the 33-parameter formula is *competitive* with the published constants of Mendoza et al., not decisively better:

- **Where it is ahead:**
  - lower mean error (1.62 % vs 2.82 %);
  - similar median (0.78 % vs 0.85 %);
  - far fewer large misses (99th percentile 11 % vs 28 %).
- **Where Mendoza et al. are ahead:**
  - few-electron ions ( $N \leq 10$ : 0.40 % vs 0.44 %);
  - the median for  $11 \leq N \leq 36$ .
- **Row by row:** the formula is closer to NIST on 51.3 % of the covered rows.
- **The 9-parameter variant:** it matches their mean on ions (2.54 % vs 2.56 %), with a larger median.
- **Model size:**
  - Mendoza et al. publish a  $19 \times 19$  matrix with 331 non-zero constants, fitted by the authors. These are counted as published numbers, not independent degrees of freedom; they are the only model-specific numbers our implementation of their IE prescription uses.
  - The formula has 33 global parameters and was also validated on held-out splits.
- **Scope:**
  - Their model also yields excitation energies and orbital properties, and is used in plasma codes. Ours gives only ionization energies.
  - The neutral-atom row lies outside the range they fitted and is shown for completeness, not as a contest.
- **Row sets:** the experimental-row value here (236 rows) differs from the all-row value of §3.1 (4.63 % on 311 rows) only because the row sets differ.

A comparison with Dirac–Fock ionization energies [30] on the same rows is left for future work.

#### 4.6 The exact first-order screening constants compared with Slater's

**Table 5.**  $\sigma_1$  for neutral-atom ground configurations, compared with Slater's  $\sigma$ . Excerpt; the full table for  $N = 1\text{--}110$  is in `results/fp_zexp_coefficients.csv`, with values for all 5847 rows in `results/fp_zexp_rows_coefficients.csv`.

| N  | configuration                                                     | removed | $E_1$ (exact)    | $\Delta E_1$ | $\sigma_1$ (ab initio) | $\sigma$ (Slater) |
|----|-------------------------------------------------------------------|---------|------------------|--------------|------------------------|-------------------|
| 2  | 1s <sup>2</sup>                                                   | 1s      | 5/8              | −0.625000    | 0.6250                 | 0.30              |
| 3  | 1s <sup>2</sup> 2s                                                | 2s      | 5965/5832        | −0.397805    | 1.5912                 | 1.70              |
| 4  | 1s <sup>2</sup> 2s <sup>2</sup>                                   | 2s      | 586373/373248    | −0.536469    | 2.1459                 | 2.05              |
| 6  | 1s <sup>2</sup> 2s <sup>2</sup> 2p <sup>2</sup> ( <sup>3</sup> P) | 2p      | 10957679/3359232 | −0.931338    | 3.7254                 | 2.75              |
| 8  | 1s <sup>2</sup> 2s <sup>2</sup> 2p <sup>4</sup>                   | 2p      | 4754911/839808   | −1.308370    | 5.2335                 | 3.45              |
| 10 | 1s <sup>2</sup> 2s <sup>2</sup> 2p <sup>6</sup>                   | 2p      | 2455271/279936   | −1.636495    | 6.5460                 | 4.15              |
| 11 | [Ne]3s                                                            | 3s      | 9.635901         | −0.865071    | 7.7856                 | 8.80              |
| 18 | [Ne]3s <sup>2</sup> 3p <sup>6</sup>                               | 3p      | 17.980333        | −1.407535    | 12.6678                | 11.25             |
| 19 | [Ar]4s                                                            | 4s      | 18.859971        | −0.879639    | 14.0742                | 16.80             |
| 29 | [Ar]3d <sup>10</sup> 4s                                           | 4s      | 39.317547        | −1.347129    | 21.5541                | 25.30             |
| 36 | [Ar]3d <sup>10</sup> 4s <sup>2</sup> 4p <sup>6</sup>              | 4p      | 50.145538        | −1.667284    | 26.6765                | 27.75             |

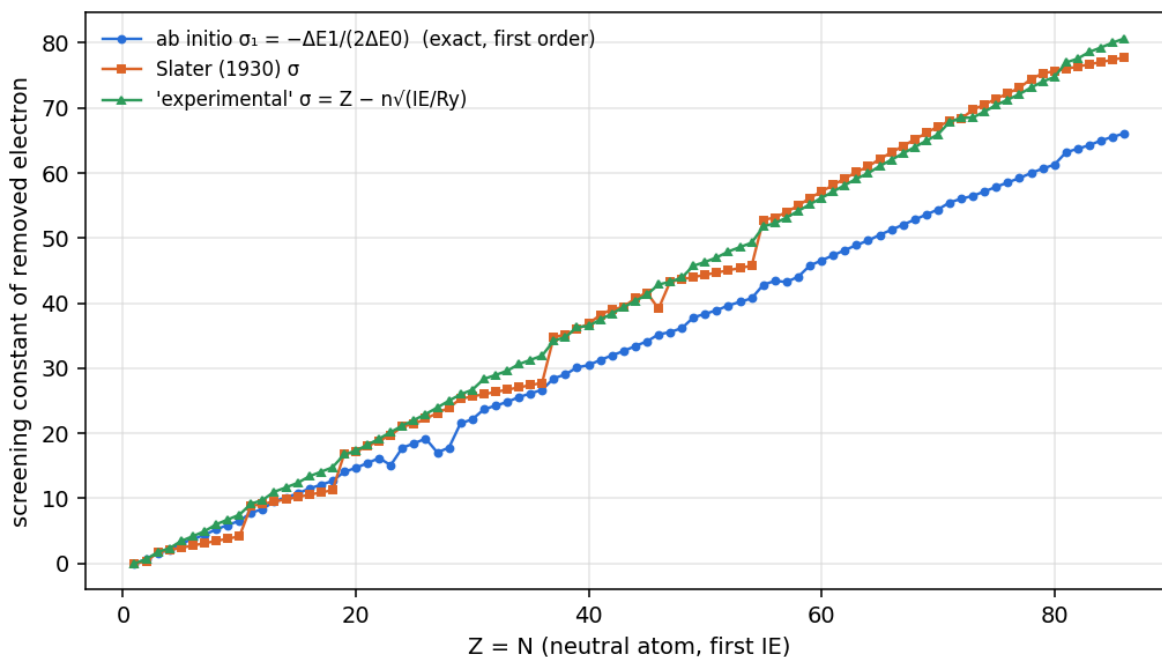
Notes:

- $E_1$  is the Hund-term single-configuration value. For  $N = 4$  (Be) the Layzer-complex value 1.5592742 is used in  $\sigma_1$ . Decimal  $E_1$  entries are exact rationals stored in the project cache.
- $\sigma_1 = -n^2\Delta E_1$ .

Slater's rules describe the neutral atom;  $\sigma_1$  is the exact screening of the bare-nucleus limit. The two disagree in a systematic way (Figure 3):

- **Inner electrons:** the hydrogenic same-shell screening is stronger than Slater's 0.35 per electron (Ne 2p: 6.55 against 4.15).
- **Valence electrons of heavy atoms:**  $\sigma_1$  is much smaller than Slater's  $\sigma$  and than the screening implied by the measured IE (Na 3s: 7.79 against Slater 8.80 and an empirical 9.16; Cs 6s: 42.8 against Slater 52.8).

This difference is exactly what the fitted remainder  $D$  supplies near neutrality.



**Figure 3.** Exact first-order screening  $\sigma_1$  against Slater's  $\sigma$  for neutral-atom configurations (existing project figure).

#### 4.7 Worked example and coverage

For the first IE of oxygen ( $1s^2 2s^2 2p^4 \rightarrow 2p^3$ , NIST 13.618 eV):

1.  $\sigma_1 = 5.23348$  and  $T = 3.3496$ .
2.  $h = 1.76652$ , so  $D = 0.71285$  and  $Z_{\text{eff}} = 2.05367$ , which lies between  $Z_a = 1$  and  $Z = 8$ .
3. The hydrogenic term is 14.3457 eV, or 14.3162 eV after relativity.
4. The Hund term is  $-0.8365$  eV.
5. IE = 13.479 eV, an error of  $-1.02\%$ .

For the first IE of nitrogen ( $1s^2 2s^2 2p^3 \rightarrow 2p^2$ , NIST 14.534 eV), a half-filled shell:

1.  $\sigma_1 = 4.37867$  and  $T = 2.9470$ .
2.  $h = 1.62133$ , so  $D = 0.63782$  and  $Z_{\text{eff}} = 1.98351$ .
3. The hydrogenic term is 13.3822 eV, or 13.3626 eV after relativity.
4. The Hund term is  $+0.8365$  eV ( $K_p(3) = +1.2$ , the opposite sign to O).
5. IE = 14.199 eV, an error of  $-2.31\%$ .

For  $\text{Mg}^{2+}$  (NIST 80.144 eV), the formula gives 78.63 eV ( $-1.88\%$ ).

**Table 6.** Case studies: first IEs of six neutral atoms, every intermediate quantity, for the three named models. Printed by `tools/audit_components.py` and `tools/case_study_table.py` (`results/case_studies.md`). The script asserts that the components reproduce the production code to  $10^{-9}$ . D is the screening remainder. The relativistic factor is  $F_{\{n,j\}} \cdot R$  for the  $\sigma_1$  models ( $R = 1$  for `pa_bound9`) and the Sommerfeld bracket for the pocket formula.

| atom<br>(removed) | model     | $\sigma_1$ | v_g<br>(same,<br>in,<br>core,<br>df,<br>out) | T        | h       | D        | Z_eff  | Ry<br>Z_eff <sup>2</sup> /n <sup>2</sup><br>(eV) | rel.<br>factor | Hund<br>(eV) | IE (eV)       | NIST<br>(eV) | error       |
|-------------------|-----------|------------|----------------------------------------------|----------|---------|----------|--------|--------------------------------------------------|----------------|--------------|---------------|--------------|-------------|
| O (2p)            | final     | 5.2335     | 3, 4, 0,<br>0, 0                             | 3.3496   | 1.7665  | 0.7128   | 2.0537 | 14.3457                                          | 0.99794        | -0.8365      | <b>13.479</b> | 13.618       | -1.02<br>%  |
| O (2p)            | pa_bound9 | 5.2335     | 3, 4, 0,<br>0, 0                             | 3.0925   | 1.7665  | 0.6173   | 2.1492 | 15.7116                                          | 1.00002        | -1.4702      | <b>14.241</b> | 13.618       | +4.58<br>%  |
| O (2p)            | pocket    | -          | 3, 4, 0,<br>0, 0                             | -        | -       | 6.1394*  | 1.8606 | 11.7747                                          | 1.00001        | -2.0735      | <b>9.701</b>  | 13.618       | -28.76<br>% |
| N (2p)            | final     | 4.3787     | 2, 4, 0,<br>0, 0                             | 2.9470   | 1.6213  | 0.6378   | 1.9835 | 13.3822                                          | 0.99853        | +0.8365      | <b>14.199</b> | 14.534       | -2.31<br>%  |
| N (2p)            | pa_bound9 | 4.3787     | 2, 4, 0,<br>0, 0                             | 2.8578   | 1.6213  | 0.5691   | 2.0522 | 14.3258                                          | 1.00001        | +1.4702      | <b>15.796</b> | 14.534       | +8.68<br>%  |
| N (2p)            | pocket    | -          | 2, 4, 0,<br>0, 0                             | -        | -       | 5.2632*  | 1.7368 | 10.2598                                          | 1.00001        | +2.0735      | <b>12.333</b> | 14.534       | -15.14<br>% |
| Na (3s)           | final     | 7.7856     | 0, 8, 2,<br>0, 0                             | 7.1802   | 2.2144  | 1.1877   | 2.0266 | 6.2091                                           | 0.99905        | 0            | <b>6.203</b>  | 5.139        | +20.70<br>% |
| Na (3s)           | pa_bound9 | 7.7856     | 0, 8, 2,<br>0, 0                             | 10.6892  | 2.2144  | 1.3219   | 1.8925 | 5.4142                                           | 1.00005        | 0            | <b>5.414</b>  | 5.139        | +5.36<br>%  |
| Na (3s)           | pocket    | -          | 0, 8, 2,<br>0, 0                             | -        | -       | 8.9608*  | 2.0392 | 6.2862                                           | 1.00006        | 0            | <b>6.287</b>  | 5.139        | +22.33<br>% |
| Ca (4s)           | final     | 14.6706    | 1, 8,<br>10, 0,<br>0                         | 24.4604  | 4.3294  | 2.8939   | 2.4356 | 5.0444                                           | 0.99659        | 0            | <b>5.027</b>  | 6.113        | -17.77<br>% |
| Ca (4s)           | pa_bound9 | 14.6706    | 1, 8,<br>10, 0,<br>0                         | 34.5745  | 4.3294  | 3.0747   | 2.2548 | 4.3233                                           | 1.00005        | 0            | <b>4.323</b>  | 6.113        | -29.28<br>% |
| Ca (4s)           | pocket    | -          | 1, 8,<br>10, 0,<br>0                         | -        | -       | 17.5935* | 2.4065 | 4.9248                                           | 1.00006        | 0            | <b>4.925</b>  | 6.113        | -19.43<br>% |
| Fe (4s)           | final     | 19.1852    | 1, 14,<br>10, 0,<br>0                        | 30.9931  | 5.8148  | 3.8109   | 3.0039 | 7.6732                                           | 0.99190        | 0            | <b>7.611</b>  | 7.902        | -3.69<br>%  |
| Fe (4s)           | pa_bound9 | 19.1852    | 1, 14,<br>10, 0,<br>0                        | 38.1569  | 5.8148  | 3.8852   | 2.9296 | 7.2983                                           | 1.00009        | 0            | <b>7.299</b>  | 7.902        | -7.64<br>%  |
| Fe (4s)           | pocket    | -          | 1, 14,<br>10, 0,<br>0                        | -        | -       | 22.8597* | 3.1403 | 8.3855                                           | 1.00011        | 0            | <b>8.386</b>  | 7.902        | +6.12<br>%  |
| Pb (6p)           | final     | 63.7240    | 1, 20,<br>60, 0,<br>0                        | 153.9342 | 17.2760 | 13.1422  | 5.1339 | 9.9612                                           | 0.79963        | +0.0465      | <b>8.012</b>  | 7.417        | +8.02<br>%  |
| Pb (6p)           | pa_bound9 | 63.7240    | 1, 20,<br>60, 0,<br>0                        | 189.5551 | 17.2760 | 13.3197  | 4.9564 | 9.2843                                           | 1.00019        | +0.0817      | <b>9.368</b>  | 7.417        | +26.31<br>% |
| Pb (6p)           | pocket    | -          | 1, 20,<br>60, 0,<br>0                        | -        | -       | 76.6038* | 5.3962 | 11.0052                                          | 1.00023        | +0.1152      | <b>11.123</b> | 7.417        | +49.97<br>% |

\*Pocket formula: total screening  $\Sigma s_g v_g + t(N-1)/(Z_a + \kappa)$ ; it has no  $\sigma_1$  and no bound.

What the table shows:

- **No s-electron extra term.** For s removal (Na, Ca, Fe) the IE is the screened Rydberg term times relativistic factors within about 1 % of unity. No model in this paper adds an s-type correction.

- **Which model is closest to NIST.**
  - The final model is the most accurate of the three on O, N, Ca, Fe and Pb.
  - pa\_bound9 is the most accurate only on Na.
  - The pocket formula is never the most accurate.
- **Ca is a clear failure of both bounded models** (−17.8 % and −29.3 %).
- **$\sigma_1$  is not the total screening.** For Pb,  $\sigma_1 = 63.72$  (Figure 3); the total screening  $\sigma_1 + D$  is 76.87.

**Coverage test.** The public API was evaluated for every  $Z = 1-118$  and every  $N = 1-Z$ : 7021 values. All are finite and positive, with no crashes. There are 22 violations of the monotonicity  $IE(Z, N-1) > IE(Z, N)$ :

- 6 at Pt–Bi, caused by 4f/5s ordering;
- 16 at Rf–Ds, where the input ground configuration jumps between neighbouring ions (e.g. Rf  $N = 68$  is  $4f^{12}6s^2$ ,  $N = 69$  is  $4f^{14}5d^1$ ).

Above  $Z = 110$  the predictions are qualitative. The 7p neutrals are too low: Og is predicted at 3.66 eV, implausibly below its lighter congener Rn (NIST 10.75 eV), and the cause is again the negative  $r_c$ .

## 5. Discussion

### 5.1 What is new and what is rediscovered

**Rediscovered, and credited as such:**

- the  $1/Z$  expansion and its exact first-order term ( $\sigma_1$  is Layzer's  $Z \rightarrow \infty$  screening constant [4], computed with textbook Slater-integral algebra [33], [34]);
- the screened hydrogenic form [1], [10], [11], and self-consistent  $(Z, N)$ -dependent screening [17], [20];
- the  $1/(Z_a + \kappa)$  remainder, which is Edlén-type isoelectronic behaviour [8];
- the one-electron Dirac, recoil, finite-size and QED corrections [31], [32];
- LSDA  $\Delta$ SCF [27], [28].

**Plausibly new, and claimed with care:**

1. **A  $\sigma_1$  table for every NIST ground configuration.**  $Z$ -expansion work [4], [5], [9] computed first-order energies for selected configurations and isoelectronic sequences. We are not aware of a table of exact first-order screening constants covering every NIST ground configuration for  $N = 1-110$ , and we offer ours as a convenient, complete tabulation rather than as a new quantity.
2. **A blind-tested hybrid formula.** A single closed form joins exact  $\sigma_1$  to a bounded fitted remainder. It is validated on all 5847 NIST successive IEs, with a pre-registered selection score and a blind extrapolation from  $Z \leq 54$  to  $Z \geq 55$ . In the SHM literature we read, we found no blind out-of-range extrapolation test.
3. **Published SHMs scored on the same rows.** The Kregar/Di Rocco model is re-implemented and scored on all 5847 rows. The constants of Mendoza et al. are scored on the 5011 rows they cover, where the formula is competitive with them rather than decisively better (§4.5).
4. **A teaching rule.** The 8-parameter pocket formula is a hand-calculable successor to Slater's rules: 4.68 % against 11.8 % on all rows, and 11.3 % against 11.6 % on blind S2.

**Not claimed:**

- a "first formula for all ionization energies", since SHMs and Dirac–Fock tables already cover all ions;
- a "first-principles formula", since only  $\sigma_1$  and the one-electron corrections are first principles and the final model has 33 fitted parameters;
- priority over machine-learning work. We did not carry out a systematic search of machine-learning studies of ionization energies, and we make no claim relative to them.

**The main lesson is methodological.** Fitting alone interpolates to about 1.6 % and extrapolates catastrophically (170 %). Exact theory alone extrapolates smoothly but is about 70 % off overall. Fixing the  $Z^2$  and  $Z$  terms by theory and confining the fit to a bounded  $O(Z^0)$  remainder gives about 2 % inside the data and 6.6 % in blind extrapolation.

## 5.2 Where the formula fails

1. **Heavy near-neutral atoms, alkaline earths and noble gases.** The blind neutral first-IE MAPE is 21.8 %, and the all-data neutral MAPE is 7.5 % (6.4 % for u35). Even with all data fitted (`results/known_failures.json`):
  - the ns<sup>2</sup> alkaline earths are too low: Ca −17.8 %, Sr −14.8 %, Ba −8.4 %;
  - Rn is too low by 24.6 %;
  - Na is too high by 20.7 % (Table 6).
  - Pb's small all-data error (+8.0 %) is partly a cancellation. The Rydberg term alone is 9.96 eV (+34 % against NIST 7.42 eV). The relativistic bracket, 0.80 because  $r_{p1/2}$  is negative, lowers it to 7.97 eV (+7.4 %), and the final IE is 8.01 eV. A term of the physically wrong sign cancels an overestimate (Table 6; `results/audit_components.md`). Valence screening in a neutral atom is an all-order, non-perturbative effect. The parameter-free LSDA  $\Delta$ SCF does better on neutral atoms (3.3 % for  $Z \leq 54$ ).
2. **An unbounded relativistic term.** The fitted  $r_c$  are all negative (−0.45 to −1.47), opposite to the direct relativistic contraction of s and  $p_{1/2}$  electrons. The bracket is not bounded, which produces the negative Lr value of the  $Z \leq 54$  fit and the low superheavy 7p IEs. A physically bounded relativistic term is the obvious next step. It must be chosen on V1/V2/S1/S3 only.
3. **f-electron removal.** This is the least accurate class: 3.5 % on all data, and about 2.0–2.5 times too high (mean 2.2) for third IEs in blind S2.
4. **Multiplet structure and rearranged configurations.** The formula has one Hund term per removed electron and no term-dependent multiplet energies. Ions whose ground configuration rearranges on ionization are described by single-configuration inputs. This is also the origin of the 22 monotonicity violations.
5. **Parameter economy.** 19 of the 33 parameters are shrunk class deviations. The bounded 9-parameter variant, pa\_bound9 (selection 2.51, blind S2 7.80 %, all data 2.90 %; parameters in Appendix A), is a reasonable alternative with fewer parameters. It was not the pre-registered winner.
6. **Reference data.** Most NIST values are labelled theoretical or semi-empirical (4617 + 919 of 5847). Agreement with them is partly agreement with other theory.

## 5.3 Why no exact closed form exists for $N \geq 2$

With  $V = \sum 1/r_{ij}$ , the many-electron Schrödinger equation does not separate.  $E(Z)$  is analytic in  $1/Z$  only up to a critical charge. For He,  $1/Z_c \approx 1.0975$ . Beyond first order, every coefficient  $E_k$  ( $k \geq 2$ ) is an infinite sum over the hydrogenic continuum, with no known closed form even for two electrons; it is known only numerically [6]. A closed formula for all ions must therefore combine exact low orders with an approximation for the rest. The question is only which approximation, and how honestly it is validated. Our answer keeps the exact orders exact and confines the fit to the remainder, under a bound that keeps it from extrapolating wildly.

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## 6. Conclusions

- **The formula.** The screened Rydberg formula gives every successive ionization energy of every element from the configuration alone, with 33 global parameters.
- **Accuracy.** 1.87 % MAPE (median 0.94 %) on 5847 NIST values; 7.5 % for neutral atoms; 0.0012 % for H-like ions.
- **Blind extrapolation.** A fit on  $Z \leq 54$  predicts  $Z = 55$ –110 with 6.60 % MAPE, under the 11.6 % of Slater's rules and the 13.6 % of a published parameter-free screened hydrogenic model. This holds because the large- $Z$  behaviour is fixed exactly by first-order perturbation theory and the fitted remainder is bounded.
- **Remaining defects.**
  - Heavy neutral atoms: in blind S2, Pb is about 1.2 eV against 7.42 and Tl about 1.7 eV against 6.11.
  - Lr is negative in the  $Z \leq 54$  fit, and Og (3.66 eV) is far below Rn.
  - Alkaline-earth neutrals are too low.
  - f-electron removal is too high in extrapolation.
  - All fitted relativistic coefficients are negative.
- **Most reusable result.** The table of exact rational first-order screening constants  $\sigma_1$  for  $N = 1$ –110.

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## Data and code availability

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All code, data and results are in the project repository, <https://github.com/AwaisSDev/Chem-Research> (folder `Np/`):

- the NIST table (`data/nist_ie.csv`) and the shared scorer (`evaluate.py`);
- the exact first-order coefficients (`results/fp_zexp_coefficients.csv`, `results/fp_zexp_rows_coefficients.csv`);
- the final model and its parameters (`models/push_a/model.py`, `models/unified/final.py`, `results/uni_final_params.json`);
- predictions for all rows (`results/uni_predictions.csv`);
- the validation script (`models/unified/validate_blind.py`);
- the literature re-implementations (`models/literature/kregar_shm.py`, `models/benchmarks/mendoza2011/`);
- every per-row model input (`results/model_inputs.csv`) and the stand-alone reference implementation (`tools/verify_from_inputs.py`), which reproduces the production predictions to  $7 \cdot 10^{-16}$ ;
- the component audit (`tools/audit_components.py`, Table 6) and the benchmark matrix with row counts in every cell (`tools/benchmark_matrix.py` → `results/benchmark_matrix.md`).

A command-line and Python interface (`ionization.py`) evaluates the formula for any  $Z \leq 118$  and  $N \leq Z$ .

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## AI-assistance disclosure

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The computations, code, analysis and a draft of this text were produced with AI agents (Anthropic Claude) working under the author's direction. The author designed and directed the study and is responsible for the content. Every number in the paper is generated by a script in the repository and can be regenerated from it. Bibliographic data were checked against Crossref. An AI system is not listed as an author.

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None.

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## Appendix A. Fitted parameters (all-data fit)

**Table A1.** Source: `results/uni_final_params.json`, which is identical to `results/pa_params.json`.

| block                                               | values                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\tau_g$ (same, in, core, df, out)                  | 0.3928, 0.7879, 2.2243, 2.5488, 10.5688                                                                                                                                                                                                                                                                       |
| $\kappa$                                            | 1.7527 (1.8027 as used in the code, which applies $ \kappa  + 0.05$ )                                                                                                                                                                                                                                         |
| $\delta\tau_c$ (19 classes, defined in Table A2)    | same_s -0.1552, same_p 0.0098, same_d 0.0004, same_f 0.1457, sn_in_p -0.1133, n1_sp_sp -0.3916, n1_sp_d 0.3009, n1_sp_f 0.2042, n2_sp_sp -0.2195, n2_sp_df -0.0622, d_near -0.6962, n1_d_d -0.2689, n1_d_f 0.3286, f_near 0.5476, n1_f_f -0.0918, out_d -0.0176, out_f 0.0173, core_sp 0.2828, core_df 0.1812 |
| $r_c$ (s, p $\frac{1}{2}$ , p $\frac{3}{2}$ , d, f) | -0.4543, -0.8129, -1.1533, -0.8426, -1.4672                                                                                                                                                                                                                                                                   |
| $x_l$ (p, d, f)                                     | 0.2049, 0.4402, 0.4612                                                                                                                                                                                                                                                                                        |

**Bounded 9-parameter variant, `pa_bound9` (all-data fit, `results/pa_params.json`, candidate `pa_bound9`).** Same equation (2.2) and inputs as the final model, with  $T = \sum_g \tau_g v_g$  (no class deviations),  $R = 1$  (no relativistic bracket), and the same  $F_{\{n,j\}}$ ,  $\mu$  and QED/FNS terms. Using these 4-decimal values instead of full precision changes no IE by more than 0.002 % (`tools/hand_recompute.py`).

| block                              | values                                  |
|------------------------------------|-----------------------------------------|
| $\tau_g$ (same, in, core, df, out) | 0.2347, 0.5971, 2.9563, 2.9515, 23.7087 |
| $\kappa$                           | 2.2091 (2.2591 as used)                 |
| $x_l$ (p, d, f)                    | 0.3602, 0.4237, 0.8215                  |

Metrics: all data 2.90 % (median 1.39 %), neutral 12.0 %, H-like 0.0012 %; V1 2.16, V2 1.95, S1 2.90, S3 3.05, selection score 2.51; blind S2 7.80 % (Table 1).

**Table A2.** Screening classes  $c$  and groups  $g$  of §2.2.  $(n, l)$  is the removed subshell and  $(n', l')$  the subshell of another electron. Each class lies inside one group. "rows" is the number of the 5847 NIST rows with  $v_c > 0$ .  $\delta\tau_c$  is from Table A1, and  $\tau_g + \delta\tau_c$  is the coefficient of one electron of that class in  $T$ . (Verified by `tools/check_grouping_rule.py`.)

| group $g$ | class $c$ | target $l$ | other electron $(n', l')$ | rows | $\delta\tau_c$ | $\tau_g + \delta\tau_c$ |
|-----------|-----------|------------|---------------------------|------|----------------|-------------------------|
| same      | same_s    | s          | $(n', l') = (n, l)$       | 534  | -0.1552        | 0.2376                  |
| same      | same_p    | p          | $(n', l') = (n, l)$       | 1713 | 0.0098         | 0.4026                  |
| same      | same_d    | d          | $(n', l') = (n, l)$       | 1720 | 0.0004         | 0.3932                  |
| same      | same_f    | f          | $(n', l') = (n, l)$       | 717  | 0.1457         | 0.5385                  |
| in        | sn_in_p   | p          | $n' = n, l' = s$          | 2073 | -0.1133        | 0.6745                  |
| in        | n1_sp_sp  | s, p       | $n' = n - 1, l' = s, p$   | 2912 | -0.3916        | 0.3963                  |
| in        | n1_sp_d   | s, p       | $n' = n - 1, l' = d$      | 1248 | 0.3009         | 1.0888                  |
| in        | n1_sp_f   | s, p       | $n' = n - 1, l' = f$      | 386  | 0.2042         | 0.9921                  |

| group g | class c  | target l | other electron (n', l')                  | rows | $\delta\tau_c$ | $\tau_g + \delta\tau_c$ |
|---------|----------|----------|------------------------------------------|------|----------------|-------------------------|
| core    | n2_sp_sp | s, p     | $n' = n - 2, l' = s, p$                  | 2083 | -0.2195        | 2.0048                  |
| core    | n2_sp_df | s, p     | $n' = n - 2, l' = d, f$                  | 668  | -0.0622        | 2.1621                  |
| core    | core_sp  | s, p     | $n' \leq n - 3$                          | 1311 | 0.2828         | 2.5070                  |
| df      | d_near   | d        | $n' = n \text{ or } n - 1, l' = s, p$    | 1938 | -0.6962        | 1.8526                  |
| df      | n1_d_d   | d        | $n' = n - 1, l' = d$                     | 1078 | -0.2689        | 2.2799                  |
| df      | n1_d_f   | d        | $n' = n - 1, l' = f$                     | 391  | 0.3286         | 2.8773                  |
| df      | f_near   | f        | $n' = n \text{ or } n - 1, l' = s, p, d$ | 778  | 0.5476         | 3.0964                  |
| df      | n1_f_f   | f        | $n' = n - 1, l' = f$                     | 201  | -0.0918        | 2.4569                  |
| df      | core_df  | d, f     | $n' \leq n - 2$                          | 2716 | 0.1812         | 2.7300                  |
| out     | out_d    | d        | $n' > n$                                 | 15   | -0.0176        | 10.5512                 |
| out     | out_f    | f        | $n' > n$                                 | 294  | 0.0173         | 10.5861                 |
| out     | sn_out   | any      | $n' = n, l' > l$                         | 0    | none           | 10.5688                 |
| out     | out_sp   | s, p     | $n' > n$                                 | 0    | none           | 10.5688                 |

**Pocket formula (8 parameters, all-data fit, results/uni\_params.json).** This is the only model called "pocket formula" in this paper.

$$Z_{\text{eff}} = Z - \sum_g s_g \nu_g - t \frac{N - 1}{Z - N + 1 + \kappa}, \quad \text{IE} = \text{Ry} \frac{Z_{\text{eff}}^2}{n^2} \left[ 1 + \frac{(Z_{\text{eff}} \alpha)^2}{n^2} \left( \frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) \right] + \text{Ry} \frac{x K_l(k)}{n^2}$$

The  $\nu_g$  are the five group counts of §2.2. The fitted values are  $s_{\text{same}}$  0.7499,  $s_{\text{in}}$  0.7514,  $s_{\text{core}}$  0.8432,  $s_{\text{df}}$  0.8408,  $s_{\text{out}}$  1.0497,  $t$  1.3725,  $\kappa$  9.815 (9.865 as used) and  $x$  0.508.